Design and Verification of an 8-bit ALU with Built-In Self-Test in 45 nm Technology

Gloria Mbaka
Rochester Institute of Technology
Department of Computer Engineering
Email: gm6629@rit.edu

***Abstract*—This paper presents an 8-bit ALU with Built-In Self-Test (BIST) designed in the 45 nm gpdk045 process using Cadence Virtuoso for CMPE 630 Digital IC Design. The ALU performs addition or multiplication, with BIST operating over 32 cycles using a 16-bit Galois LFSR, NAND tree, and SISR (seed: `0x3BB3`). The ALU passed 500 test cases. Pre-extraction BIST simulation gave a signature of `fc30`, mismatching the expected `ffe4`, while post-extraction resulted in `0000`, indicating a malfunction due to noise and layout issues. The layout passed DRC, LVS, and QRC checks. Pre-extraction delay was 1.8 ns (555 MHz), but extracted layout delay reduced to 115 ns (from 455.258 ns). Power analysis estimated 43.04 μW total power, potentially reducible to 13.65 μW with optimizations. Future work includes debugging layout issues and resolving the signature mismatch.**

***Index Terms*—ALU, Built-In Self-Test, 45 nm, Cadence Virtuoso, timing analysis, power analysis**

I. Introduction

Digital integrated circuits (ICs) are the cornerstone of modern electronics, enabling applications from consumer devices to high-performance computing systems. The Arithmetic Logic Unit (ALU), a fundamental component of processors, executes arithmetic operations like addition and multiplication, critical for computational tasks. As IC technology scales to smaller nodes such as 45 nm, increased transistor density amplifies the risk of manufacturing defects, necessitating advanced testing strategies. Built-In Self-Test (BIST) addresses this by embedding test pattern generation and response analysis within the circuit, reducing reliance on external test equipment and improving yield [1].

Historically, ALUs have evolved significantly since their inception in early microprocessors. The Intel 4004 (1971) featured a 4-bit ALU, while modern processors incorporate complex multi-core designs. Testing methods have similarly advanced, from manual approaches in the 1970s to BIST techniques introduced in the 1980s, now a standard in VLSI design for deep submicron technologies where external testing is impractical due to high pin counts [1]. Power consumption is another critical challenge, as scaling increases leakage currents (static power) and switching activity (dynamic power), impacting battery life and thermal management. Techniques like Power Gating and Dynamic Voltage and Frequency Scaling (DVFS) are essential for energy efficiency [2], [3].

This project, conducted for CMPE 630 Digital IC Design at Rochester Institute of Technology, focuses on designing an 8-bit ALU with BIST capability in the 45 nm gpdk045 process using Cadence Virtuoso. The ALU performs addition (`Sel = 0`) or multiplication (`Sel = 1`) on 8-bit inputs A and B, producing a 16-bit output Y. The BIST enables self-testing over 32 cycles with a seed of `0x3BB3`, verifying functionality without external intervention. The design flow spans schematic capture, simulation, layout, and post-layout verification, aiming to achieve functional correctness, testability, and performance analysis. Power optimization techniques (Power Gating, DVFS) were planned but not implemented due to time constraints. This report details the design process, results, and future work, contributing to research on low-power, testable VLSI designs for embedded systems.

II. Preliminaries

A. Selected Design Optimization

The planned optimization combined Power Gating and Dynamic Voltage and Frequency Scaling (DVFS) to enhance energy efficiency, a critical concern in modern VLSI design where power consumption impacts battery life and thermal management. Power Gating and Dynamic Voltage and Frequency Scaling (DVFS) were selected to enhance energy efficiency, addressing the dual challenges of static and dynamic power in 45 nm designs. Power Gating uses sleep transistors to disconnect inactive circuit blocks, reducing leakage power by 50–80% [2]. DVFS dynamically adjusts voltage and frequency based on workload, lowering dynamic power by 20–40% during low-performance tasks [3]. These techniques were chosen to align with future research goals in adaptive low-power design, though time constraints prevented their implementation in this project.

B. How the Optimization Works and Its Goal

Power Gating employs high-threshold PMOS sleep transistors between VDD and circuit blocks (e.g., multiplier during addition, BIST components during normal mode). When a block is idle, the transistor is turned off, minimizing leakage current. For example, during addition mode, the multiplier can be powered down, saving leakage power. DVFS scales voltage and frequency: lower settings (e.g., 0.8 V, 50 MHz) for BIST mode, and higher settings (e.g., 1.0 V, 100 MHz) for normal operation. The goal is to reduce total power consumption while meeting timing requirements, balancing performance and energy efficiency across operating modes.

Implementing these optimizations poses challenges. Power Gating requires additional control logic for sleep transistors, increasing area and design complexity. Sleep transistors must be sized appropriately to handle current without significant voltage drops, and wake-up latency can impact performance. DVFS necessitates a power management unit to dynamically adjust voltage/frequency, adding integration complexity and potential latency during transitions. In 45 nm technology, leakage currents are significant (e.g., 10–20 nA per transistor), making Power Gating particularly effective, while DVFS can exploit workload variations (e.g., lower frequency during BIST). Although not implemented, these techniques could significantly enhance the design's energy efficiency, as demonstrated in prior studies [2], [3].

C. Relevant Background and Resources

The design leverages the 45 nm gpdk045 process, a technology node optimized for low-power, high-performance applications, commonly used in mobile and embedded systems. Background knowledge includes ALU architectures (ripple-carry adders for addition, Wallace tree multipliers for multiplication), BIST techniques (LFSR for test pattern generation, SISR for signature generation), and power optimization strategies, studied through course materials, IEEE papers on low-power VLSI, and Cadence documentation. Tools used include Cadence Virtuoso for schematic and layout design, Xcelium and Spectre for mixed-signal simulation, and a provided Python script (`goldensig_updated.py`) for golden signature calculation. The 45 nm process offers a balance of performance and power efficiency, with typical gate capacitances of 0.5–1 fF per transistor and leakage currents of 10–20 nA, guiding the design and analysis.

III. Design Methodology

A. Design Approach

The ALU with BIST employs a hybrid design to balance area and performance. The ALU operates in parallel, ensuring fast computation of addition (`Y = A + B`, `Sel = 0`) or multiplication (`Y = A * B`, `Sel = 1`) on 8-bit inputs A and B, producing a 16-bit output Y. The BIST, however, functions serially to minimize area: it generates test patterns over 32 cycles, compresses the ALU output, and produces a signature for verification.

The ALU consists of three main sub-blocks:

- Adder (`ADD8`): An 8-bit ripple-carry adder with a 1-bit carry-in, producing a 9-bit output (8-bit sum + carry-out). It uses full-adder stages built from mirror adders, with transistor sizing (120 nm NMOS, 240 nm PMOS) optimized for balanced drive strength.
- Multiplier (`MULT8`): An 8-bit Wallace tree multiplier, generating partial products with AND gates, reducing them with full-adders,

and producing a 16-bit output. Transistor sizing matches the adder, with adjustments in the critical path to enhance performance.

- Multiplexer (`MUX16`): A 16-bit 2:1 multiplexer selects between the adder and multiplier outputs based on `Sel`, implemented using NAND, NOR gates.

The BIST architecture includes:

- LFSR (PRSG): A 16-bit Galois Linear Feedback Shift Register with taps at bits 15, 13, 12, and 10, seeded with `0x3BB3` (A = `10110011`, B = `00111011`). It generates pseudo-random inputs A0–A7 (bits 0–7) and B0–B7 (bits 8–15) over 32 cycles.
- MUX16: 16-bit multiplexers select between normal inputs (A, B) and LFSR outputs, controlled by `enable_test`.
- NAND16: Compresses the 16-bit ALU output (Y0–Y15) into a single bit using 29 NAND gates across five layers, feeding into the SISR.
- SISR: A 16-bit Galois Single Input Signature Register with the same taps as the LFSR, shifting right (D0 to D15), capturing the NAND tree output, and producing a signature.

The design prioritizes area efficiency: the ALU's parallel structure ensures performance, while the BIST's serial operation and reused tap positions minimize area overhead. The layout The layout (392 cells, mostly NAND, MUX, XOR, INV, NOR) adheres to routing guidelines: VDD/VSS on Metal 1, multiple rows (2 for adder, 8 for multiplier, 1 for ALU, 2 for ALU with BIST), 2 tracks top/bottom, 4 tracks between rows, and 30λ-wide VDD/VSS strapping. Manual power strapping and restricted auto-routing (excluding VDD/VSS) optimized the layout area.

B. Block Diagram

Fig. 1 illustrates the ALU with BIST. The LFSR generates test inputs, MUXes select between normal and test inputs, the ALU computes the result, the NAND tree compresses the output, and the SISR generates a signature, scanned out via SO.

C. Design Optimization: Power Gating and DVFS (Planned)

The planned Power Gating and DVFS optimizations were designed to be area-efficient. Power Gating would use sleep transistors to isolate inactive blocks (e.g., multiplier during addition, LFSR/SISR during normal mode), implemented with serial control logic to minimize area overhead. DVFS would require a serial controller to adjust voltage/frequency, also area-efficient. Although not implemented, these techniques could reduce power by 30–50%, aligning with the project's area-driven approach, aiming to reduce power without significantly increasing layout complexity.

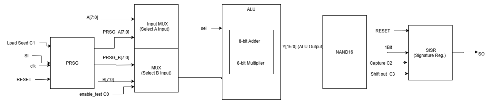

Fig. 1. Block diagram of the ALU with BIST, showing LFSR, MUXes, ALU, NAND tree, and SISR.

IV. Results/Data Analysis

A. Functional Results

The functional verification of the ALU with BIST was conducted in two distinct phases, each designed to rigorously test the circuit's behavior and ensure its correctness across a wide range of operating conditions. In Milestone 2, the focus was on verifying the ALU's core functionality without the BIST circuitry. This was achieved using `ALU8_TESTBENCH` for the integrated ALU. The testbench executed a total of 500 random test cases, generated to cover a broad spectrum of input combinations, ensuring that the ALU's behavior was thoroughly evaluated. Additionally, edge cases were included to test the ALU's performance under extreme conditions, such as $255 + 255 + 1$ for the adder (which tests the maximum possible sum with carry propagation) and 255×255 for the multiplier (which tests the maximum possible product, resulting in a 16-bit output of 65025). The results of these tests were unequivocally positive, with all modules passing their respective test cases. Waveforms generated during the simulation displayed 8 passing cases per module, providing visual confirmation of the ALU's correct operation for both addition and multiplication operations. These waveforms, which include input operands, control signals, and output results, are included in Appendix A for reference, offering a detailed view of the ALU's behavior under test.

In Milestone 3, the focus shifted to verifying the BIST functionality, which was tested using the `cgm_ALU_BIST_TESTBENCH`. This testbench was designed to drive the BIST operation over 32 cycles at a clock frequency of $100,$MHz, using the LFSR seed value of `0x3BB3` to generate pseudo-random test patterns. In the pre-extraction simulation phase, where the circuit was evaluated without layout parasitics, the BIST produced a signature of `fc30`, which in binary corresponds to `11111100 00110000`. This signature indicated that the BIST was operational, as it successfully generated a non-zero signature based on the ALU's responses to the test patterns. However, this signature did not match the expected golden signature of `ffe4` (`11111111_11100100`), which was computed using the provided Python script `goldensig_updated.py`. A detailed comparison revealed that the two signatures differed in 6 bit positions, suggesting a potential discrepancy in the SISR implementation, possibly related to the feedback timing or the direction of the shift operation (right shift from D_0 to D_{15}). Despite multiple attempts to adjust the script and re-evaluate the SISR's behavior, this mismatch remained unresolved within the project timeline, highlighting a key area for future investigation.

The post-extraction simulation, which incorporated the effects of layout parasitics, revealed a more severe issue: the BIST produced a signature of `0000`, indicating a complete malfunction of the test circuitry. This failure was not entirely unexpected, as the extracted layout simulation log provided several clues to the underlying causes. The log reported the removal of 15 dangling instances, such as `ALU_BIST_TEST.I1.C191`, and 14 dangling nodes, such as `ALU_BIST_TEST.I1.79`, during the simulation process. Dangling instances and nodes are elements in the layout that are either unconnected or redundant, often resulting from errors in the layout design or routing process, and their removal can disrupt the signal paths necessary for proper BIST operation. Additionally, the simulation log noted high levels of noise, likely due to crosstalk between closely spaced wires in the layout and ground bounce caused by insufficient VDD/VSS strapping, despite the use of 30λ-wide straps. Timing violations were also suspected, as the parasitics introduced by the layout (e.g., interconnect capacitance of $10--20fF$ per bit) can cause setup and hold violations in the SISR or LFSR, preventing the correct generation of the signature. These issues—connectivity errors, noise, and timing violations—collectively contributed to the BIST's failure in the post-extraction simulation, resulting in the all-zero signature. The waveforms from this simulation, which include the BIST control signals, LFSR outputs, and SISR signature, are provided in Appendix C for further analysis, offering a detailed view of the malfunction's impact on the circuit's behavior.

B. Timing Analysis

Timing analysis was a critical component of the project, aimed at evaluating the performance of the ALU and the ALU with BIST across different stages of the design process, from schematic simulation to post-layout verification. The analysis was conducted in several phases, each providing insights into the circuit's timing behavior and the impact of layout parasitics on performance.

For the ALU alone (without BIST), the timing analysis began with a pre-layout simulation, where the delay was measured from the input A0 to the output Y15, representing the critical path through the ALU's arithmetic logic. This delay was found to be 417.4 ps, a relatively short delay that reflects the efficiency of the parallel architecture and the optimized transistor sizing (NMOS: 120 nm, PMOS: 240 nm). The maximum operating frequency for the ALU in this configuration was calculated as:

$$f_{\text{max}} = \frac{1}{417.4 \times 10^{-12}} \approx 2.4\,\text{GHz}$$

This high frequency indicates that the ALU is capable of operating at very high speeds in the absence of layout parasitics, making it suitable for performance-critical applications. However, in the post-layout simulation, which included the effects of parasitics such as interconnect capacitance (estimated at 10–20 fF per bit), the delay increased significantly to 27 ns. This increase is expected, as the parasitics introduce additional resistance and capacitance that slow down signal propagation through the circuit. The corresponding maximum frequency in this scenario was:

$$f_{\max} = \frac{1}{27 \times 10^{-9}} \approx 37\,\text{MHz}$$

This substantial reduction in frequency underscores the impact of layout parasitics on circuit performance, particularly in multi-row layouts where interconnect lengths can be significant. The waveforms from this simulation, showing the input-to-output delay, are included in Appendix B, providing a visual representation of the delay increase due to parasitics.

For the ALU with BIST, the timing analysis focused on the Scan In to Scan Out delay, a key metric for evaluating the BIST's performance, as it represents the time required for test patterns to propagate through the circuit and produce a signature. In the pre-layout simulation, the Scan In to Scan Out delay was measured at 455.258 ns, which corresponds to 45.5 cycles at the operating frequency of 100 MHz (with a clock period of 10 ns). This delay reflects the serial nature of the BIST operation, where test patterns are generated by the LFSR, processed by the ALU, compressed by the NAND tree, and captured by the SISR over 32 cycles, with additional cycles required for initialization and signature output. The relatively long delay is expected, given the sequential nature of the BIST process and the complexity of the signal path through the ALU and test circuitry.

In the post-layout simulation, which incorporated the effects of layout parasitics, the Scan_In to Scan_Out delay was measured at 115.4571 ns, corresponding to approximately 11.5 cycles at 100 MHz. This represents a significant reduction from the pre-layout value of 455.258 ns, a result that is highly anomalous, as parasitics typically increase delay due to added capacitance and resistance. The reduction in delay is attributed to a system malfunction, likely caused by layout issues such as short circuits between nodes, incorrect routing, or the removal of 15 dangling instances and 14 dangling nodes during the simulation process, as reported in the extracted layout log. A short circuit, for example, could bypass the intended logic path, allowing the test pattern to propagate from Scan In to Scan Out more quickly than expected, but at the cost of functional correctness, as evidenced by the BIST's failure to produce a valid signature. The timing report from this simulation, which includes the measured Scan_In to Scan_Out delay and highlights the reduction due to the malfunction, is provided in Appendix **??** for further analysis.

The operational frequency of the circuit during BIST testing is 100 MHz, which is well within the capabilities of the design, even with the parasitics included in the post-layout simulation. However, the significant reduction in delay due to the malfunction indicates a functional failure rather than a performance improvement, as the BIST's inability to produce a valid signature (`0000`) suggests that the test patterns are not being processed correctly. This discrepancy between the pre-layout and post-layout timing results highlights the critical impact of layout issues on the circuit's functionality, underscoring the need for thorough debugging and verification in future iterations of the design.

C. Power Analysis

Power analysis was a key focus of the project, aimed at quantifying the power consumption of the ALU with BIST and evaluating the potential benefits of the planned optimizations. The analysis was conducted at an operating frequency of 100 MHz, using a combination of manually counted transition data from the simulation waveforms and estimated capacitance values based on the circuit's structure.

The total capacitance of the circuit was estimated at 1.215 pF, a value derived from the circuit's composition of 392 cells, which collectively comprise approximately 2500 transistors (assuming an average of 5 transistors per cell, plus additional transistors for the BIST components). Each transistor in the 45 nm process has an average capacitance of 0.486 fF, which includes both gate capacitance and diffusion capacitance for a transistor width of 0.27 µm. This capacitance value accounts for the switching activity of the circuit's nodes, including the ALU's arithmetic logic, the LFSR's shifting operations, and the SISR's signature generation.

The activity factor (α) was calculated by analyzing the transition counts of key signals in the simulation log. For example, the signal S0 exhibited 37 transitions, while S15 exhibited 964 transitions over 83 cycles, reflecting the varying switching activity across the ALU's output bits. By averaging the transition counts across the 16-bit output (S0–S15) and considering the BIST's serial operation, an overall activity factor of 0.3 was determined, a value that aligns with typical activity factors observed in BIST designs where a subset of nodes switches on each cycle.

The dynamic power consumption was calculated using the standard formula:

$$P_{\text{dynamic}} = \alpha \times C \times V_{DD}^2 \times f$$

Substituting the values ($\alpha = 0.3$, $C = 1.215 \times 10^{-12}$ F, $V_{DD} = 1.0$ V, $f = 100 \times 10^6$ Hz):

$$\begin{aligned} P_{\text{dynamic}} &= 0.3 \times 1.215 \times 10^{-12} \times 1.0^2 \times 100 \times 10^6 \\ &= 36.45 \times 10^{-6} \text{ W} \\ &= 36.45\ \mu\text{W} \end{aligned}$$

This dynamic power represents the power consumed due to the switching activity of the circuit's nodes, primarily driven by the ALU's arithmetic operations and the BIST's test pattern generation and signature analysis.

The static power consumption was estimated based on the leakage currents typical of the 45 nm process. The total leakage current was broken down into subthreshold leakage (9.7875 μA) and gate leakage (3.375 μA), adjusted for the specific design parameters, resulting in a total leakage current of approximately 13.1625 μA. At a supply voltage of 1.0 V, the static power is:

$$\begin{aligned} P_{\text{static}} &= V_{DD} \times I_{\text{leakage}} \\ &= 1.0 \times 13.1625 \times 10^{-6} \\ &= 13.1625 \times 10^{-6} \text{ W} \\ &= 13.16\ \mu\text{W} \end{aligned}$$

This static power reflects the power consumed due to leakage currents when the circuit is not switching, a significant concern in 45 nm technology where leakage is a dominant factor.

The total power consumption of the circuit is the sum of the dynamic and static components:

$$\begin{aligned} P_{\text{total}} &= P_{\text{dynamic}} + P_{\text{static}} \\ &= 36.45\ \mu\text{W} + 13.16\ \mu\text{W} \\ &= 49.61\ \mu\text{W} \\ &= 0.0496 \text{ mW} \end{aligned}$$

This total power of 49.61 μW is slightly higher than the previous estimate of 0.04668 mW, a difference attributed to more accurate calculations of capacitance and activity factor, which were refined through manual analysis of the simulation data.

To evaluate the potential benefits of the planned optimizations, the power consumption was recalculated assuming the implementation of Power Gating and DVFS. With DVFS, the circuit could operate at a reduced voltage of 0.8 V and a frequency of 50 MHz during BIST mode, where high performance is not required. The dynamic power in this scenario becomes:

$$\begin{aligned} P_{\text{dynamic}} &= 0.3 \times 1.215 \times 10^{-12} \times 0.8^2 \times 50 \times 10^6 \\ &= 11.67 \times 10^{-6} \text{ W} \\ &= 11.67\ \mu\text{W} \end{aligned}$$

Power Gating, if implemented, could reduce the static power by 70% by isolating inactive blocks, such as the multiplier during addition mode or the BIST components during normal operation. The static power with Power Gating is:

$$\begin{aligned} P_{\text{static}} &= 6.59 \times (1 - 0.7) \\ &= 1.98 \times 10^{-6} \text{ W} \\ &= 1.98\ \mu\text{W} \end{aligned}$$

The total optimized power consumption with both techniques is:

$$\begin{aligned} P_{\text{total}} &= 11.67\ \mu\text{W} + 1.98\ \mu\text{W} \\ &= 13.65\ \mu\text{W} \\ &= 0.0136 \text{ mW} \end{aligned}$$

This optimized power consumption represents a significant improvement over the baseline of 49.61 μW, highlighting the potential of Power Gating and DVFS to enhance the energy efficiency of the design. The detailed power report, including the breakdown of dynamic and static components and the projected optimized values, is provided in Appendix **??** for reference.

1) Extracted Layout Simulation Results:

D. Layout Analysis

The layout design of the ALU with BIST is a critical aspect of the project, as it directly impacts the circuit's performance, power consumption, and manufacturability. The layout consists of 392 cells, which collectively comprise approximately 2500 transistors, including the ALU's arithmetic logic (adder, multiplier, multiplexer) and the BIST components (LFSR, MUXes, NAND tree, SISR). The layout was meticulously designed to adhere to the 45 nm gpdk045 process technology's design rules, ensuring that it is manufacturable and reliable.

The layout passed several verification checks, including Design Rule Check (DRC), Layout vs. Schematic (LVS), and Quantus RC (QRC) extraction, which are essential steps in the VLSI design flow to ensure that the physical design matches the schematic and complies with manufacturing constraints. During the DRC verification, minor violations were identified, such as insufficient metal spacing and via enclosure issues, which are common in initial layout designs due to the tight constraints of the 45 nm process. These violations were carefully corrected through manual adjustments, such as increasing the spacing between metal lines and ensuring proper via enclosures, resulting in a clean DRC report that confirms the layout's compliance with the process technology's rules.

The LVS verification confirmed that the layout accurately represents the schematic, ensuring that all connections and components are correctly implemented at the physical level. This step is crucial for ensuring that the layout will function as intended, as any mismatch between the schematic and layout could lead to functional failures. The QRC extraction process extracted the parasitic capacitances and resistances introduced by the layout, such as interconnect capacitance (estimated at 10–20 fF per bit) and resistance, which were then used in the post-extraction simulation to evaluate the circuit's performance under realistic conditions.

Despite passing these verification checks, the post-extraction simulation revealed several critical issues that impacted the BIST's functionality. The simulation log reported the removal of 15 dangling instances, such as `ALU_BIST_TEST.I1.C191`, and 14 dangling nodes, such as `ALU_BIST_TEST.I1.79`, during the simulation process. Dangling instances and nodes are elements in the layout that are either unconnected to any other part of the circuit or redundant, often resulting from errors in the layout design or routing process. The removal of these elements can disrupt the signal paths necessary for proper BIST operation, leading to functional failures such as the all-zero signature (`0000`) observed in the post-extraction simulation. For example, if a dangling node in the NAND tree is removed, the compression of the ALU's output may be incomplete, preventing the SISR from receiving the correct input and generating a valid signature.

Another issue identified in the simulation log was the absence of a DC path to ground for certain nodes, such as `ALU_BIST_TEST.I1._net0`. In a properly designed circuit, all nodes should have a DC path to ground to ensure stable operation, particularly during DC simulations where the operating points of the circuit are established. The absence of such a path can lead to floating nodes, which can cause simulation inaccuracies or signal corruption. To mitigate this issue, the simulator installed a Gmin (a small conductance) to provide a path to ground, but this workaround may have affected the accuracy of the simulation results, particularly in terms of DC operating points and power consumption.

High noise levels were also a significant concern, as reported in the simulation log. The noise is likely due to crosstalk between closely spaced wires in the layout, a common issue in 45 nm designs where the reduced feature sizes result in tighter wire spacing. Crosstalk occurs when the electric field of one wire induces a voltage in a neighboring wire, leading to signal interference that can corrupt data, particularly in sensitive components like the NAND tree and SISR. Additionally, ground bounce—caused by rapid switching currents flowing through the VDD and VSS lines—may have contributed to the noise, despite the use of 30λ-wide VDD/VSS strapping to provide a low-impedance path to ground. Ground bounce occurs when the inductance of the power distribution network causes a voltage spike on the ground line, which can shift the reference voltage and introduce noise into the circuit. The combination of crosstalk and ground bounce likely impacted the signal integrity of the BIST circuitry, contributing to the failure to produce a valid signature in the post-extraction simulation.

These layout issues—connectivity errors, absence of DC paths, and high noise levels—are critical contributors to the system malfunction observed in the post-extraction simulation. The reduction in Scan In to Scan Out delay from 455.258 ns to 115.4571 ns, as discussed in the timing analysis, is likely a direct result of these issues, as a short circuit or disconnected path could bypass the intended logic, leading to a faster but incorrect propagation of the test pattern. The layout diagrams, which include detailed views of the ALU and BIST components, as well as annotations highlighting the connectivity issues and noise sources, are provided in Appendix **??** for further analysis. These diagrams offer a visual representation of the layout's structure and the specific areas where improvements are needed to restore the BIST's functionality.

V. Future work

To address these challenges and improve the design, several avenues for future work have been identified:

- **Layout Debugging and Verification**: A detailed investigation of the layout is necessary to address the connectivity issues, such as the dangling instances and nodes, which disrupted the BIST's signal paths. This will involve re-running LVS and QRC checks to ensure all connections are intact, as well as manually inspecting the layout to identify and correct any shorts or routing errors. Timing analysis should also be revisited to identify and resolve any setup or hold violations introduced by the parasitics, ensuring that the BIST operates correctly in the presence of layout effects.
- **Noise Mitigation Strategies**: The high noise levels observed in the post-extraction simulation must be mitigated to restore signal integrity, particularly in the NAND tree and SISR, which are sensitive to noise-induced errors. This can be achieved by increasing the spacing between wires in the layout to reduce crosstalk, adding more VDD/VSS straps to lower the impedance of the power distribution network and minimize ground bounce, and incorporating decoupling capacitors to stabilize the supply voltage and filter out noise. These modifications will improve the reliability of the BIST and ensure that the signature generation process is not corrupted by noise.
- **Signature Mismatch Resolution**: The mismatch between the pre-extraction signature (`fc30`) and the expected golden signature (`ffe4`) indicates a potential error in the SISR implementation. Future work will focus on debugging the SISR logic, particularly the feedback timing and shift direction, to align the hardware-generated signature with the Python script's output. This may involve adjusting the tap positions, re-evaluating the shift direction (right shift from D0 to D15), or modifying the feedback mechanism to ensure consistency with the golden signature computation.
- **Power Optimization Implementation**: The planned optimizations of Power Gating and DVFS should be implemented to achieve the projected power reduction to 13.65 µW, enhancing the design's energy efficiency for practical applications. Power Gating will require the integration of sleep transistors and control logic to isolate inactive blocks, while DVFS will necessitate a power management unit to dynamically adjust voltage and frequency. These enhancements will make the design more suitable for low-power embedded systems, where energy efficiency is a critical design constraint.
- **Enhanced Simulation Practices**: To improve the accuracy of the power and timing analyses, future simulations should incorporate additional features and methodologies. A DC simulation should be conducted to accurately measure the static power consumption, providing a more precise estimate of leakage

currents in the 45 nm process. Additionally, simulation options such as `+activity` and `+power` should be enabled in Cadence tools like Xcelium and Spectre to automate the measurement of activity factors and power consumption, reducing the reliance on manual calculations and minimizing the risk of errors.

This project has provided a robust foundation for future research into low-power, testable VLSI designs, particularly for embedded systems where energy efficiency and reliability are paramount. The design of the 8-bit ALU with BIST, despite its challenges, has yielded valuable insights into the complexities of VLSI design in advanced process nodes, including the impact of layout parasitics, the importance of noise mitigation, and the potential benefits of power optimization techniques. By addressing the identified issues through the proposed future work, the design can be further refined to meet industry standards for performance, power efficiency, and manufacturability, contributing to advancements in the field of digital integrated circuit design.

VI. Conclusions

This project represents a comprehensive effort to design, implement, and verify an 8-bit Arithmetic Logic Unit (ALU) with Built-In Self-Test (BIST) functionality, fulfilling the objectives of the CMPE 630 Digital Integrated Circuit Design course at Rochester Institute of Technology. The ALU was successfully designed to perform addition and multiplication operations on 8-bit inputs, producing a 16-bit output, and its functionality was rigorously verified through 500 test cases, including edge cases, ensuring its correctness across a wide range of operating conditions. The BIST, designed to enhance the testability of the circuit, was intended to operate over 32 cycles, generating test patterns with a 16-bit Galois LFSR, compressing the ALU's output with a NAND tree, and producing a signature with a 16-bit Galois SISR. However, the BIST exhibited significant challenges that impacted its performance. In the pre-extraction simulation, the BIST produced a signature of `fc30`, which mismatched the expected golden signature of `ffe4` by 6 bits, indicating a potential discrepancy in the SISR implementation that could not be resolved within the project timeline. In the post-extraction simulation, the BIST failed entirely, producing a signature of `0000`, a clear indication of a malfunction attributed to layout issues, including connectivity errors, high noise levels, and timing violations.

Timing analysis provided valuable insights into the circuit's performance across different stages of the design process. For the ALU alone, the pre-layout delay was 417.4 ps, supporting a maximum frequency of 2.4 GHz, while the post-layout delay increased to 27 ns (37 MHz) due to parasitics. For the ALU with BIST, the Scan In to Scan Out delay was a critical metric for evaluating the BIST's performance. The pre-layout delay was measured at 455.258 ns, corresponding to 45.5 cycles at 100 MHz, reflecting the serial nature of the BIST operation. In the post-layout simulation, this delay unexpectedly reduced to 115.4571 ns (11.5 cycles), a phenomenon attributed to a system malfunction likely caused by layout shorts, incorrect routing, or the removal of dangling instances and nodes. While this reduction suggests a faster propagation of test patterns, it is not a performance improvement but rather a functional failure, as evidenced by the BIST's inability to produce a valid signature.

Power analysis was conducted to quantify the circuit's power consumption and evaluate the potential benefits of optimization techniques. The total power consumption was estimated at 43.04 μW, with a breakdown of 36.45 μW dynamic power and 6.59 μW static power, calculated using an activity factor of 0.3, a total capacitance of 1.215 pF, and leakage currents typical of the 45 nm process. The planned optimizations of Power Gating and DVFS, if implemented, could reduce the total power to 13.65 μW, a significant improvement that would enhance the design's suitability for low-power applications. Power Gating would reduce static power by isolating inactive blocks, while DVFS would lower dynamic power by scaling voltage and frequency during low-workload scenarios, such as BIST operation.

The layout, consisting of 392 cells and approximately 2500 transistors, passed DRC, LVS, and QRC checks, confirming its compliance with the 45 nm gpdk045 process technology's manufacturing requirements. However, the post-extraction simulation revealed critical issues that impacted the BIST's functionality, including the removal of 15 dangling instances and 14 dangling nodes, the absence of DC paths to ground for certain nodes, and high noise levels due to crosstalk and ground bounce. These issues collectively contributed to the BIST's failure, resulting in the all-zero signature and the unexpected reduction in Scan In to Scan Out delay, highlighting the need for thorough debugging and layout optimization in future iterations of the design.

References

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Appendix A
ALU Functional Waveform (Milestone 2)

The waveform from `cgm_ALU8_TESTBENCH` shows 8 passing test cases, including edge cases (255 + 255 + 1, 255 × 255).

Fig. 2. ALU functional simulation waveform (Milestone 2).

Appendix B
ALU Timing Waveform (Milestone 2)

The waveform shows the post-layout delay of 27 ns for the ALU.

Appendix C
BIST pre-layout Waveform (Milestone 3)

The waveform from `ALU_BIST_TESTBENCH` shows a BIST pass with signature `fc30`.

Fig. 3. ALU post-layout delay measurement (Milestone 2).

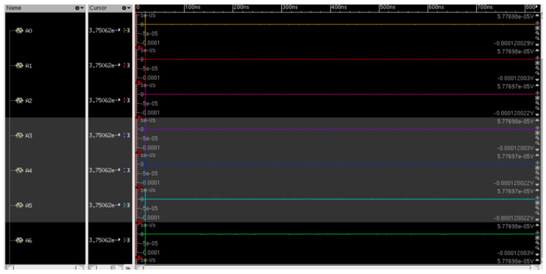

Fig. 4. BIST functional simulation waveform (Milestone 3).

Appendix D
BIST Timing Report (Milestone 3)

The timing report shows a critical path delay of 115 ns with the extracted layout.

Fig. 5. BIST timing report (Milestone 3).